8086 Rotate Instructions

1. Overview of Rotate Instructions

Concept

Rotate instructions move bits in a **circular fashion**. Unlike shift instructions, **no bits are lost**. Bits rotated out from one end re-enter at the other end.

Rules / Notes

General Syntax

```
ROL destination, count
ROR destination, count
RCL destination, count
RCR destination, count
```

Count can be:

- Immediate value = 1
- CL register

2. ROL (Rotate Left)

Concept

ROL rotates all bits to the left. The MSB is copied into the LSB **and** into the Carry Flag (CF).

Rules / Notes

Operation

- MSB $\rightarrow$ LSB
- MSB $\rightarrow$ CF

Example

Bit-Level Example

$$1001\ 0110 \xrightarrow{ROL} 0010\ 1101$$

Example

```
MOV AL, 96H     ; 1001 0110
ROL AL, 1
; AL = 2DH, CF = 1
```

Exam Tip

ROL is often used in cryptography and checksum algorithms.

3. ROR (Rotate Right)

Concept

ROR rotates all bits to the right. The LSB is copied into the MSB **and** into CF.

Rules / Notes

Operation

- LSB → MSB
- LSB → CF

Example

Bit-Level Example

$$1001\ 0110 \xrightarrow{ROR} 0100\ 1011$$

Example

```
MOV AL, 96H
ROR AL, 1
; AL = 4BH, CF = 0
```

Exam Tip

ROR is commonly used for bit-field manipulation.

4. RCL (Rotate through Carry Left)

Concept

RCL rotates bits to the left **through the Carry Flag**. CF becomes part of the rotation.

Rules / Notes

Operation

- MSB $\to$ CF
- CF $\to$ LSB

Example

Bit-Level Example (CF = 1)

$$\text{CF} = 1,\ 1001\ 0110 \xrightarrow{RCL} 0010\ 1101$$

Example

```
STC                ; CF = 1
MOV AL, 96H
RCL AL, 1
; AL = 2DH, CF = 1
```

Exam Tip

RCL is useful for multi-precision arithmetic.

5. RCR (Rotate through Carry Right)

Concept

RCR rotates bits to the right **through the Carry Flag**. CF participates in rotation.

Rules / Notes

Operation

- LSB $\to$ CF
- CF $\to$ MSB

Example

Bit-Level Example (CF = 0)

$$\text{CF} = 0,\ 1001\ 0110 \xrightarrow{RCR} 0100\ 1011$$

Example

```
CLC              ; CF = 0
MOV AL, 96H
RCR AL, 1
; AL = 4BH, CF = 0
```

Exam Tip

RCR can be used to rotate multi-byte values across registers.

6. Flags Affected by Rotate Instructions

Rules / Notes

- **CF**: receives last rotated bit
- **OF**: defined only for 1-bit rotate
- **SF, ZF, PF: not affected**

Exam Tip

For rotate count > 1, the Overflow Flag is undefined. This is a common exam question.

Quick Revision (One Look)

- **ROL**: rotate left (no CF input)
- **ROR**: rotate right (no CF input)
- **RCL**: rotate left through CF
- **RCR**: rotate right through CF
- No data loss (unlike shift)
- Rotate count = 1 or CL